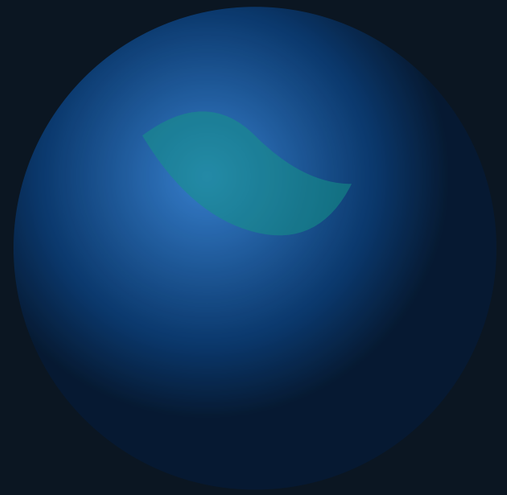
WEEK 1 · DAY 1

# Colab Bootcamp + Your First Time Series

Today you open Google Colab, load real satellite-style data and draw your first time-series chart. 🛰️



NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

# Three moves. That's all.



## Open Colab

your Python lives in the browser



## Load the data

3 years of Ka-band availability



## Draw the story

find the rainy-season dip yourself

⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 60 min hands-on · 🏁 20 min wrap-up

QUICK QUESTION

# Did NEPA bring light yesterday? 💡

Here is one house, hour by hour. Yellow = light. Dark = “NEPA has taken light.”




### 🕐 A time

every hour, 00 → 23

### ⚡ A value

light on (1) or off (0)

### = a time series

You have been collecting data all your life. “Up NEPA!” is a data point. 😄

PART 1

# Colab Bootcamp 💻

Your free Python computer that lives inside Google Chrome.

WHAT IS COLAB?

# Your laptop just shows the screen. Google's computer does the work.



**Your laptop**

only needs Chrome



**Google's computer**

already has Python, pandas, matplotlib, scikit-learn



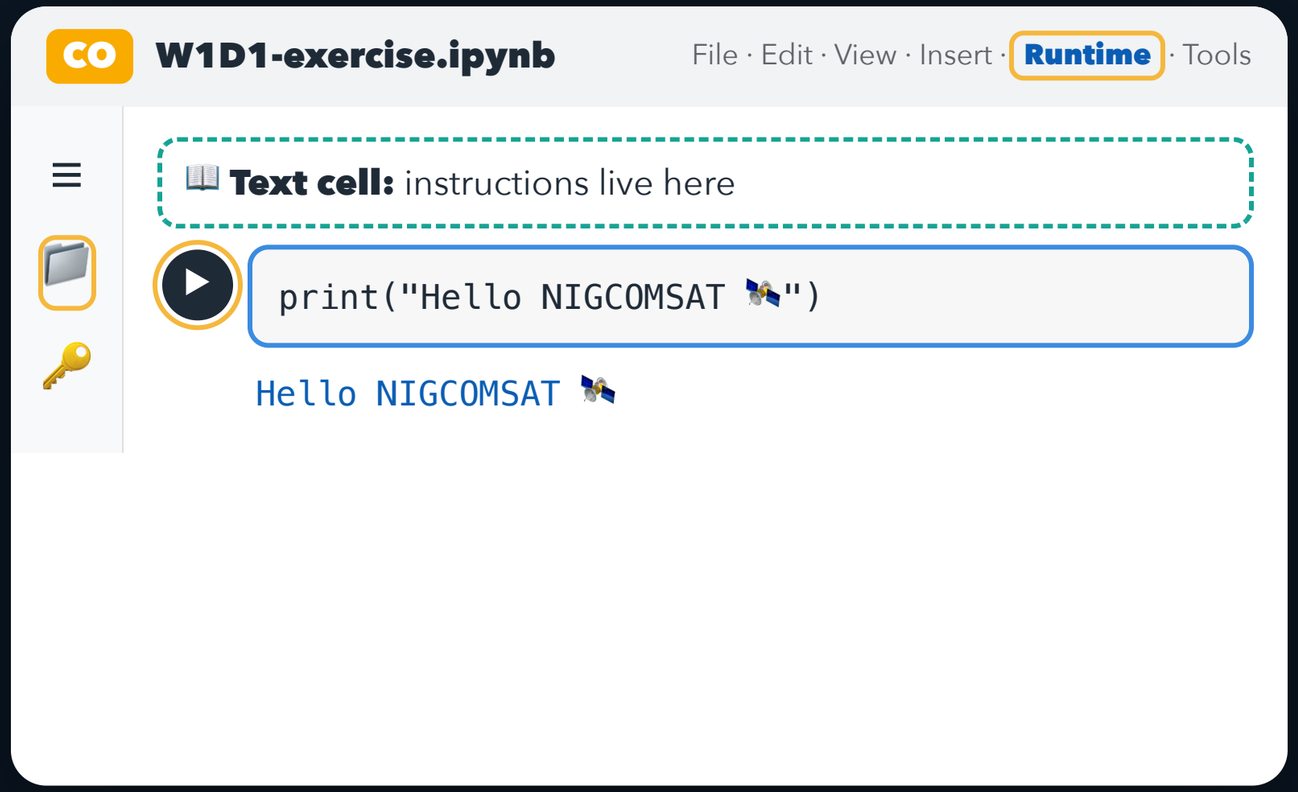
**Your answers**

tables + charts, back on your screen

Like a cyber café computer… **but free, and it never runs out of fuel.** ⛽😄

FIND YOUR WAY AROUND

# The Colab screen in 5 parts




1. ▶ **Run button**: runs one cell. Shortcut: **Shift + Enter**
2. **Code cell** (blue box): Python goes here
3. **Text cell** (green box): reading, not running
4. 📁 **Files panel**: data you download or upload
5. **Runtime menu**: restart and “Run all”

COLAB'S 3 GOLDEN RULES

# Follow these and Colab won't embarrass you



### 1. Save a copy first

File → **Save a copy in Drive**. Otherwise your work vanishes when you close the tab.



### 2. Top to bottom

Run cells **in order** with Shift + Enter. Skipping a cell is like skipping a step in a recipe.



### 3. Confused? Restart

Runtime → **Restart session and run all**. It fixes 80% of weird problems.

**Rule 1 is the big one. Do it now, before anything else.**

WHY DO I GET NAMEERROR?

# Colab has a short memory 🧠

What Colab remembers right now

```
hours_in_year = 8760
```

```
ka = [1096 rows]
```

```
TIME_COLUMN = "timestamp"
```

### 🔄 Restart session…

…or leave Colab idle for too long. The memory is wiped. 🧹

```
NameError: name 'ka' is not defined
```

Translation: **“I don't remember. Run the cells above again.”**

THE AI BUTTON (GEMINI) IN COLAB

# AI can help you. It must never see restricted data.



## OK to paste

This course's training data (it is synthetic, made for teaching)
Your own code and error messages
“Explain this line to me”



## Never paste

Real NIGCOMSAT telemetry, RF records, network logs
Customer or staff details
Anything your unit has not cleared

Not sure? **Describe the shape** (“a table of hourly voltages”), **don't paste the data.**

PART 2

# What is a time series? ⏱️

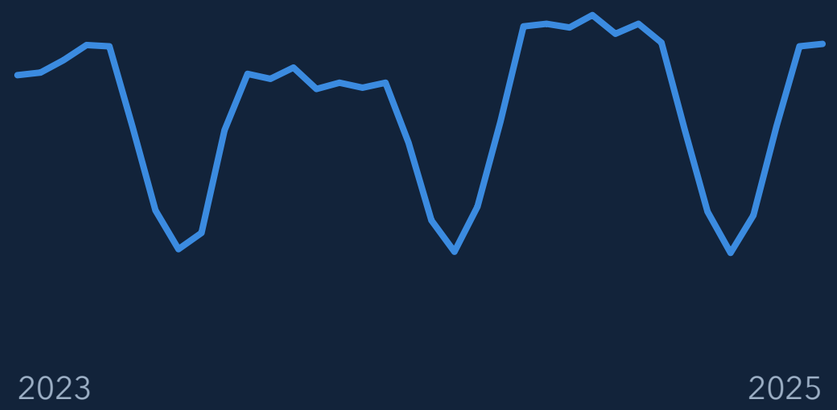
One idea. Once you get it, the whole course makes sense.

THE ONE BIG IDEA

# A time series is a table where the ORDER of the rows carries meaning

## ✅ In time order

Ka-band availability, 36 months




You can see the yearly rainy-season dip.

## 🔀 Same numbers, shuffled

exactly the same 36 values




The story is gone. Order **is** information.

ANATOMY

# Every time series has 3 things

🕐

## A timestamp

when it happened

```
2025-08-01T21:00
```

🔢

## A value

what was measured

```
battery_soc_pct = 77.3
```

🥁

## A rhythm (cadence)

how often it is recorded

**hourly** **daily** **weekly** **monthly**

Like checking your phone: some people check **every minute**, some **once a day**. Same life, different cadence. 📱

MEET THE STAR OF THE COURSE

# NigComSat-1R

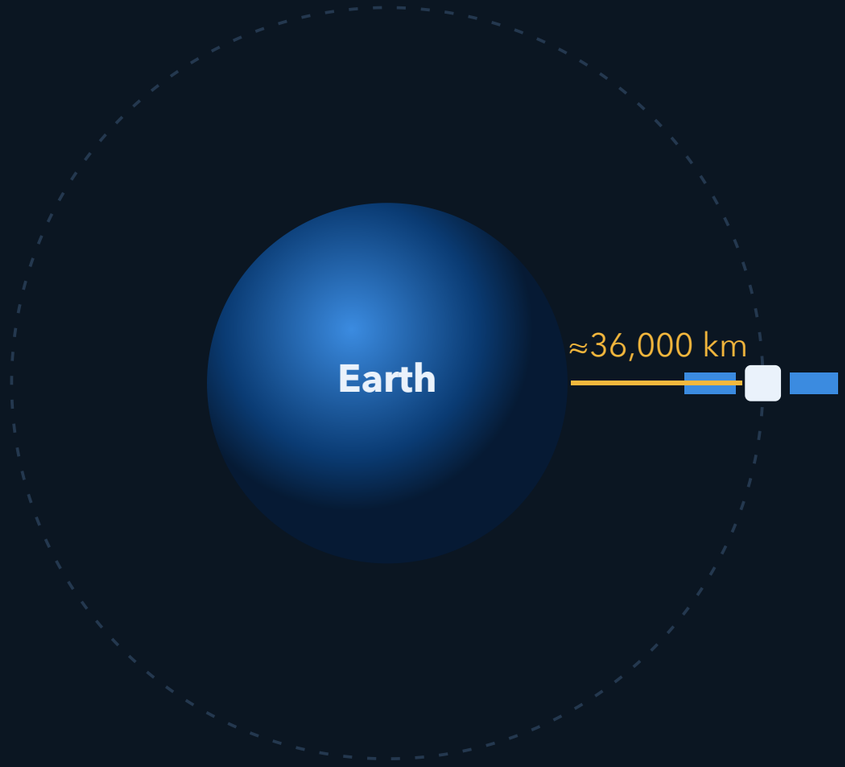
Nigeria's communications satellite, parked over the equator at **42.5° East**.

**28**
active transponders

**36,000**
km above the Earth (approx.)

**C·Ku·Ka·L**
four frequency bands

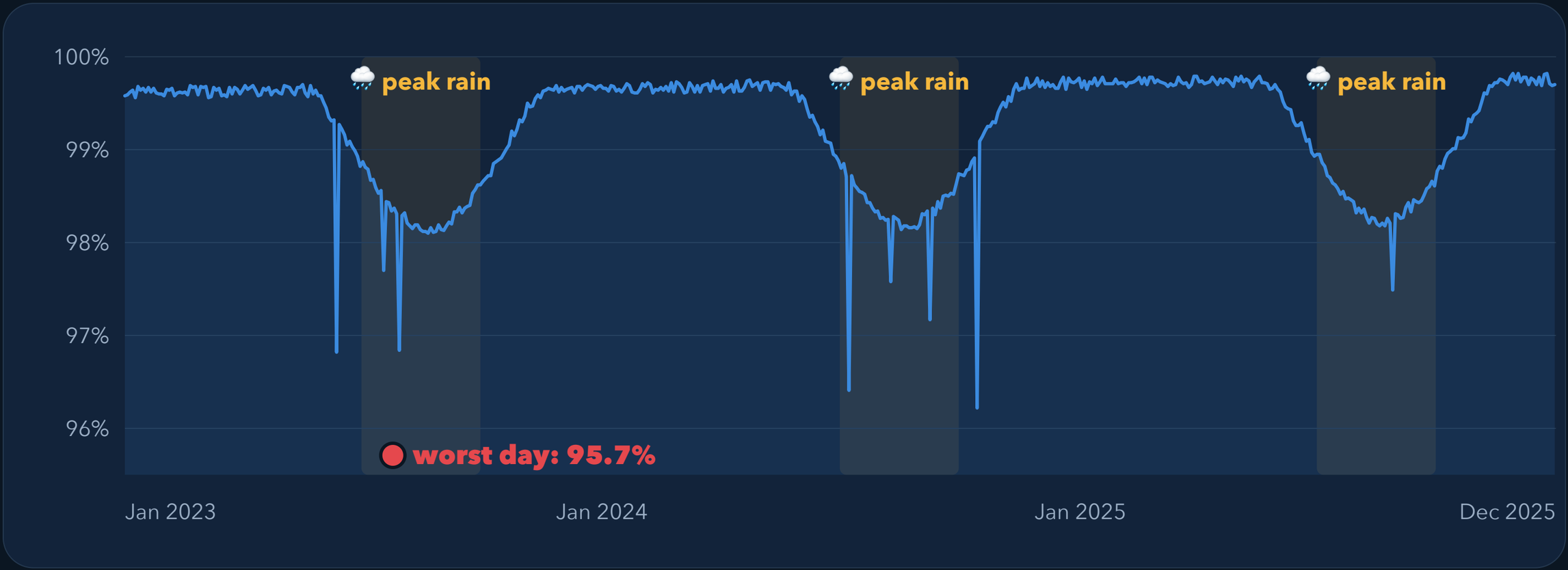
**15 yrs**
design life (launched 2011)




geostationary orbit: it stays over the same spot

TODAY'S DATASET · KA_BAND_AVAILABILITY_DAILY.CSV

# 3 years of the Ka-band spot beam over Nigeria




1,096 days

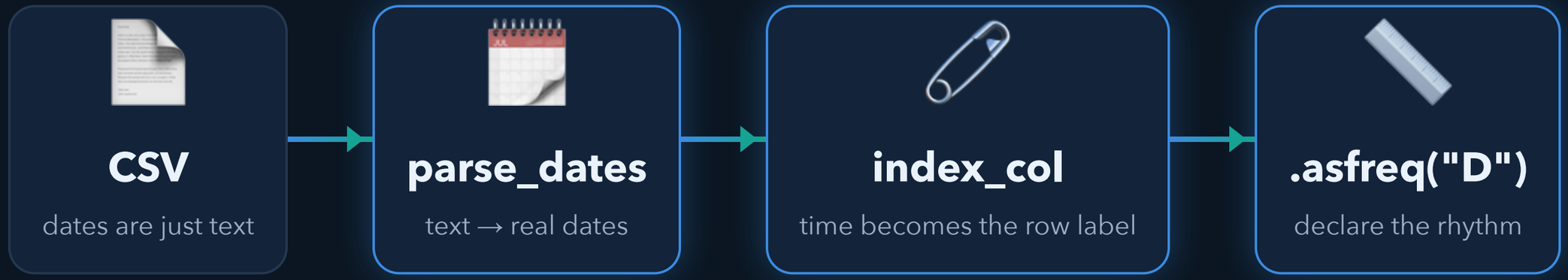
no gaps

dips every rainy season = rain fade

synthetic teaching data

TEACH PANDAS ABOUT TIME

# The 3-line recipe you will use every day




```
ka = pd.read_csv(data_path("ka_band_availability_daily"),
                 parse_dates=["timestamp"],      # 1. text -> dates
                 index_col="timestamp")          # 2. time is the row label
ka = ka.asfreq("D")                              # 3. one row per day, gaps show up
```

WHY .ASFREQ()?

# Missing days must show up, not hide

Export from a system: day 6, 7 and 8 never arrived. Can you see it?

| 1 | 2 | 3 | 4 | 5 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|

11 rows. Looks fine. **It is lying.** 🙈

After `.asfreq("D")`: one slot for every day

| 1 | 2 | 3 | 4 | 5 | NaN | NaN | NaN | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

14 rows. The 3 missing days appear as **NaN** (“Not a Number” = empty). Now you can count them. ✅

BEFORE YOU TRUST ANY NEW DATA

# Inspect it like a tokunbo car 🚗🔍

### Rows?

how many readings

### Range?

first and last date

### Gaps?

missing periods

### Duplicates?

same time twice

### Missing?

empty cells (NaN)

You check the engine, tyres and papers **before** you pay. Check these 5 **before** you analyse. 💸

TIME ZONES

# Satellites speak UTC. Lagos speaks WAT.



**21:00 UTC**

satellite telemetry: eclipse time



**22:00 WAT**

Lagos / Abuja clock: UTC + 1, all year

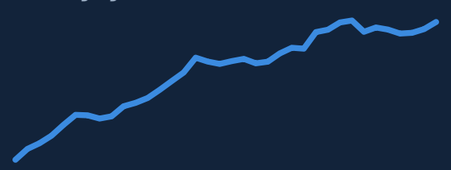
**Rule:** use ONE time zone before joining two tables. (Nigeria has no daylight saving 🙌)

SNEAK PEEK OF WEEK 2

# Every series is made of 4 characters

## Trend

The long direction. Like Lagos: more people every year.



transponder use, 2023→2025

## Seasonality

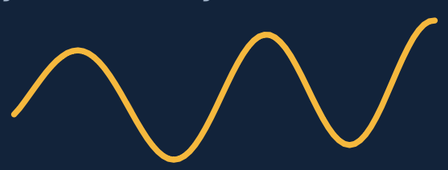
Repeats on a fixed calendar. Rainy season, every year.



Ka-band availability, monthly

## Cycle

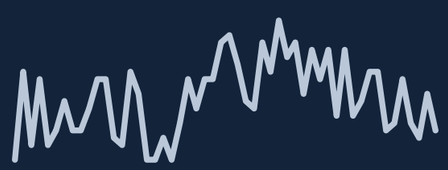
Ups and downs with NO fixed length. Boom years, slow years.



illustration

## Noise

Random wobble. 38 tickets vs 41 tickets = same week.



NOC tickets per week, 2025

YOUR TURN · 60 MINUTES

# Open the exercise notebook 👩🏾‍💻

1. Click **Open in Colab** on W1D1-exercise
2. **File → Save a copy in Drive** 💾
3. Run **STEP 0**, then go top to bottom
4. Replace every ____, run, get the ✅



**12 checks to pass**

Part A: Colab warm-up
Part B: your first time series
🌶️ Bonus: the pepper task

Keep the **rule book** open next to you.

TODAY IN 3 LINES

# What you now know



**Colab:** save a copy, run top to bottom, restart when confused.



**Time series:** a table where **order matters**: timestamp + value + cadence.



**pandas recipe:** parse_dates → index_col → .asfreq() so gaps can't hide.

Next time 👉 **Charts That Don't Lie**: why the *average* hid a whole rain outage. 🌧️